



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to AKC SERVICE STATION

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

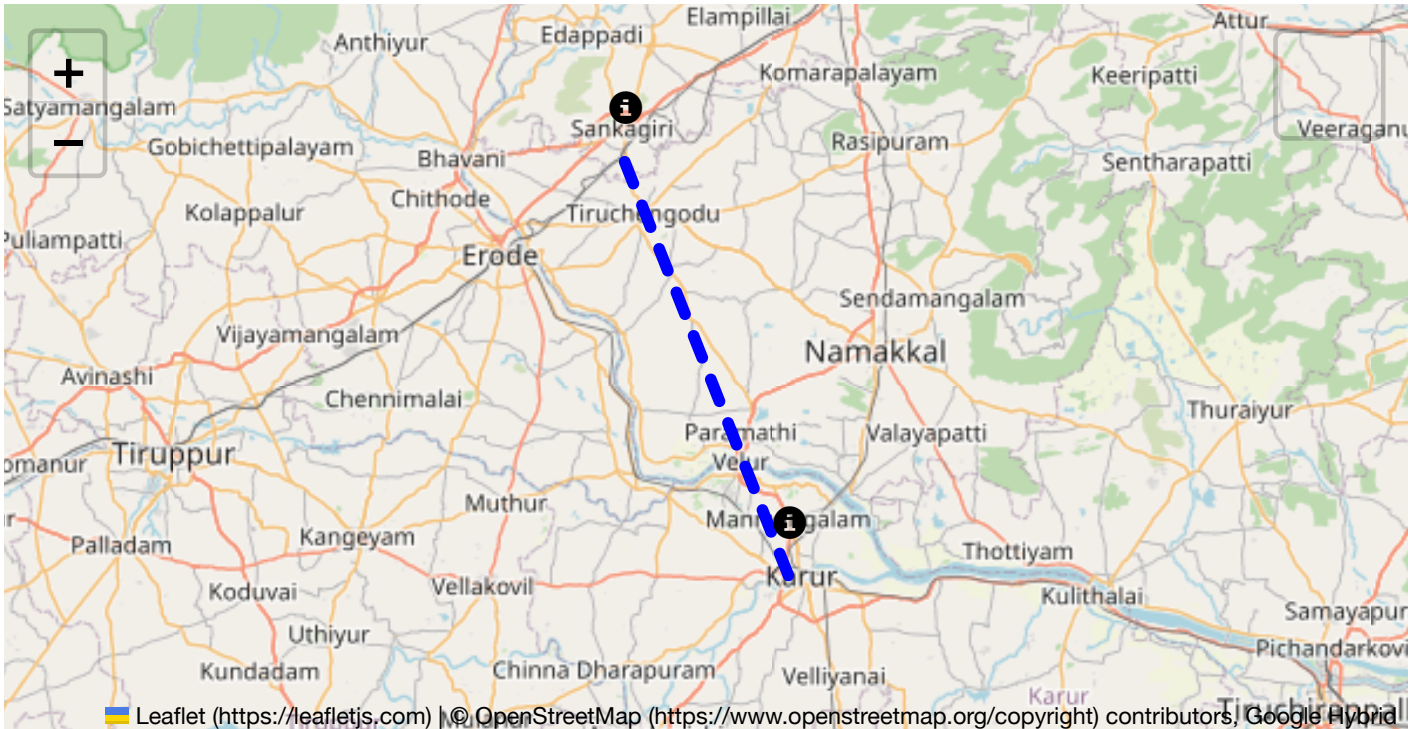
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

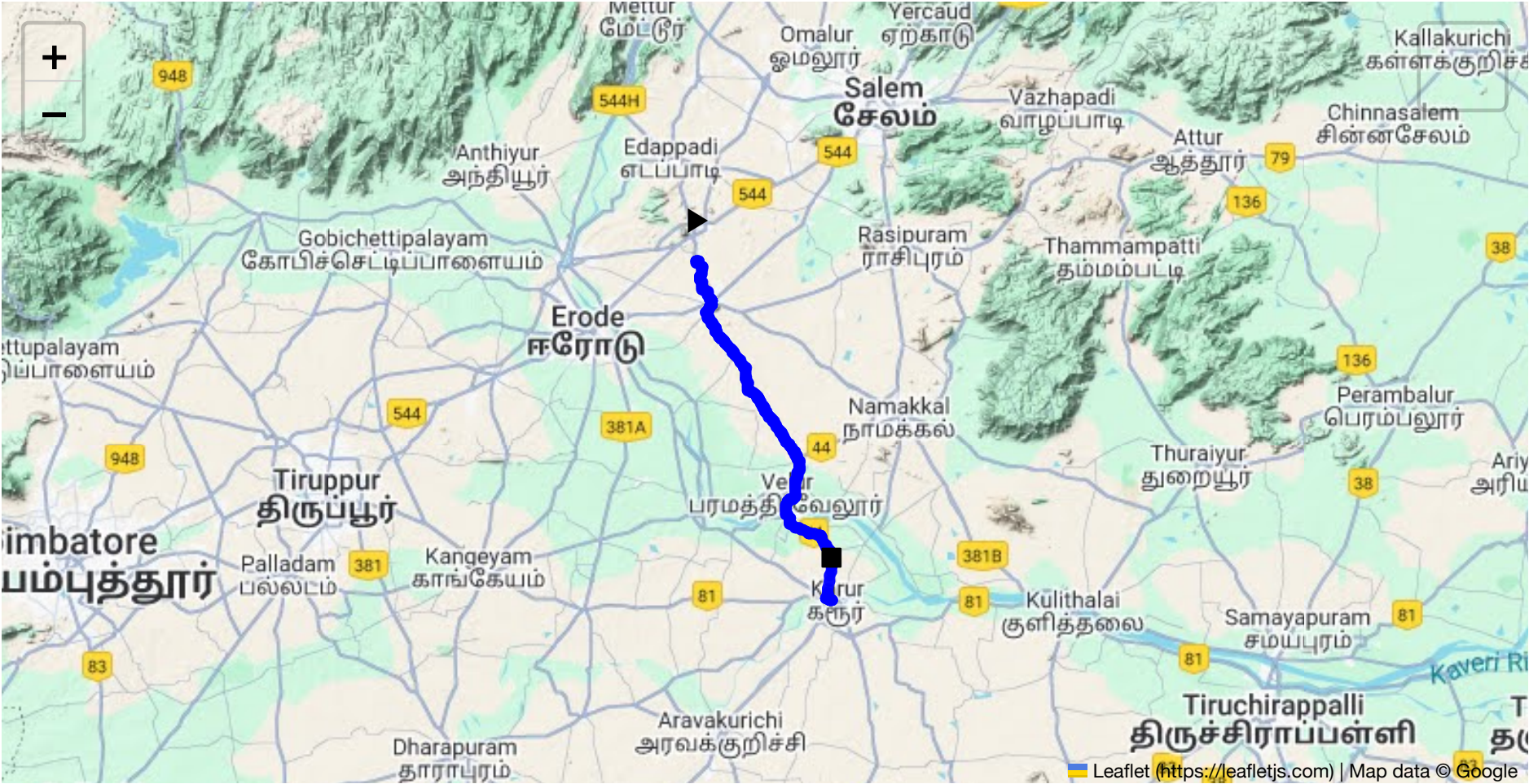
Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 65.36 km
Estimated Duration: 1.3 hours
Adjusted Duration (Heavy Vehicle): 1.6 hours

Start: (11.4381, 77.8734)
End: (10.961079, 78.065604)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route from Sangagiri, Tamil Nadu 637301 to Karur 639002 is approximately 65.36 kilometers. Starting from Sangagiri, the route typically follows major state highways and national highways, passing through a mix of rural and semi-urban regions until reaching Karur.

2. Typical Weather Conditions and Potential Weather-Related Hazards

Tamil Nadu generally experiences a tropical climate. The region can have:

- **Hot summers (March to May):** High temperatures can lead to road heating, causing potential tire blowouts.
- **Monsoon season (June to September):** Heavy rainfall might result in slippery roads, flooding, and reduced visibility.
- **Mild winters (December to February):** Typically no extreme weather, but morning fog can impact visibility.

3. Analysis of Traffic Patterns

- **Morning and Evening Peak Hours:** Expect higher traffic volumes from 8:00 AM to 10:00 AM and 5:00 PM to 7:00 PM, especially in towns and near urban centers.
- **Congestion-Prone Areas:** Urban fringes and market areas might experience bottlenecks, particularly near town centers along the route.

4. Assessment of Road Quality and Infrastructure

- **National Highways:** Generally well-maintained with signage and markings, though caution is advised around ongoing construction.
- **Rural Roads:** May vary in condition, with potential for potholes and uneven surfaces particularly after monsoon season.

5. Suggestions for Alternative Routes for Emergencies

- **Alternate Route A:** Consider bypass roads around major towns to avoid urban congestion if necessary.
- **Alternate Route B:** Use secondary roads parallel to highways that might offer quicker passage during major roadblocks or accidents.

6. Summary of Local Regulations Affecting Hazardous Material Transport

- **Regulatory Compliance:** Ensure adherence to rules for transporting hazardous materials, including clearly marked signage on trucks, adherence to speed limits, and holding necessary permits.
- **Prohibited Zones:** Verify any restricted areas near populated or environmentally sensitive regions.

7. Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials

- **Accident Reports:** Casual incidents of spills or traffic accidents involving heavy vehicles have occurred historically, mostly attributed to over-speeding or poor road conditions.
- **Key Lessons:** Rigorous adherence to safety standards, including regular vehicle maintenance, is essential.

8. Environmental Considerations and Sensitive Areas

- **Protected Areas:** Avoid sensitive zones around any wildlife sanctuaries or areas prone to erosion.
- **Local Ecology:** Be mindful of the local ecosystem, especially during the dry season when vegetation is prone to damage from road dust.

9. Analysis of Communication Coverage

- **Strong Coverage:** Urban areas along the route typically have good mobile network coverage.
- **Potential Dead Zones:** Rural stretches might have weaker signals; carrying a satellite phone for emergencies is advisable.

10. Estimated Emergency Response Times for Different Route Segments

- **Urban Areas:** Shorter response times of approximately 15-20 minutes due to proximity to facilities.
- **Rural Areas:** Possible delays, with response times extending to 30-45 minutes.

12. Overall Summary of Risk Assessment

- **Medium Risk Level:** While the route is generally safe with proper precautions, variables such as weather, road conditions, and urban congestion necessitate caution.

- Key Recommendations:** Regularly check weather forecasts, maintain communication equipment, ensure trucks meet safety standards, and drivers are trained for handling hazardous materials. Always have contingency plans for alternate routes and thoroughly understand local traffic laws and regulations concerning hazardous transport.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Roundabout	High	10.96304, 78.06131	15 KM/Hr
2	U-Turn	High	10.9607089, 78.0679609	10 KM/Hr
3	Turn	Medium	11.43822, 77.87348	30 KM/Hr
4	Turn	High	11.43968, 77.87345	15 KM/Hr
5	Blind Spot	Blind Spot	11.44029, 77.87544	10 KM/Hr
6	Turn	High	11.37868, 77.89498	15 KM/Hr
7	Turn	High	11.37850, 77.89453	15 KM/Hr
8	Turn	Medium	11.16764, 78.01565	30 KM/Hr
9	Turn	High	11.15573, 78.02176	15 KM/Hr
10	Turn	High	11.14545, 78.02208	15 KM/Hr
11	Turn	High	11.14480, 78.02291	15 KM/Hr
12	Blind Spot	Blind Spot	10.96071, 78.06796	10 KM/Hr
13	Turn	High	10.96066, 78.06791	15 KM/Hr
14	Blind Spot	Blind Spot	10.96123, 78.06585	10 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	T.C.A Hospital Tiruchengode	11.3791885, 77.8965774	30 km/h	Medium
6	clinic	Kongu Nursing Home	11.3783065, 77.8961134	30 km/h	Medium
7	hospital	Soorya Multispecialty Hospital	11.3786429, 77.8931912	30 km/h	Medium
8	hospital	Tiruchengode Government Hospital	11.37645, 77.89426	30 km/h	Medium
9	hospital	Krishna Hospital, Namakkal	11.3754811, 77.8931817	30 km/h	Medium
10	hospital	Tirukumaran Hospitals	11.3782118, 77.8914933	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
13	police	கரட்டுப்பாளையம் காவல் நிலையம்	11.357586, 77.8922539	30 km/h	Medium
14	hospital	Goverment Hospital, Nallur	11.2679698, 77.946863	30 km/h	Medium
16	hospital	Goverment Hospital	11.156647, 78.0213586	30 km/h	Medium
18	hospital	Government Hospital, Manmangalam	11.0275504, 78.0646437	30 km/h	Medium
19	hospital	Abishek Ortho Center	10.9628788, 78.0600715	30 km/h	Medium
20	hospital	Abishek Ortho Center, Karur	10.9624485, 78.059568	30 km/h	Medium
21	clinic	Senthil Nursing Home, Karur	10.9620158, 78.0643985	30 km/h	Medium
22	hospital	Apollo Hospital	10.96044, 78.064468	30 km/h	Medium
23	hospital	Murugesan Hospital	10.9619995, 78.0643134	30 km/h	Medium
24	hospital	Senthil's Emergency Care	10.9619826, 78.0643823	30 km/h	Medium
25	hospital	Kumaran Hospital	10.9619213, 78.0644924	30 km/h	Medium
26	hospital	Vasan Eye Care Hospital, Karur	10.9608117, 78.0665959	30 km/h	Medium
27	clinic	V.C Nursing Home	10.9605156, 78.0663214	30 km/h	Medium
28	hospital	Amaravathi Hospital	10.9602192, 78.0684534	30 km/h	Medium
29	hospital	Karuvurar Siddha Hospital	10.9617419, 78.0689621	30 km/h	Medium
30	hospital	Arasan Eye Hospital	10.961065, 78.067647	30 km/h	Medium
31	hospital	K R G Hospital	10.9633415, 78.0684828	30 km/h	Medium
32	hospital	Loganathan Dental Care Hospital	10.962963, 78.068556	30 km/h	Medium
33	hospital	Gugan Hospital	10.9601178, 78.0691481	30 km/h	Medium
34	hospital	Poongothai Hospital	10.9619987, 78.0690593	30 km/h	Medium
35	hospital	Ranjith Child Hospital	10.9622369, 78.0690826	30 km/h	Medium
36	clinic	Renu Nursing Home	10.9594975, 78.0704542	30 km/h	Medium
37	hospital	Jeyammal Hospital	10.9633541, 78.070428	30 km/h	Medium
38	hospital	Nesi Eye Care Hospital	10.9612243, 78.0704182	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
11	school	KSR Educational institution	11.3772013, 77.8908807	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
12	school	MDV School	11.3719843, 77.8915244	30 km/h	Medium
15	school	Sivabakkiam Muthusamy Matriculation Higher Secondary School	11.1788177, 78.0061696	30 km/h	Medium
17	school	Goverment Higher Secondary School	11.1511724, 78.0200723	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.38395, 77.89480



Risk Type: Roundabout
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.96304, 78.06131



Risk Type: U-Turn

Risk Level: High

Speed Limit: 10 KM/Hr

Coordinates: 10.9607089, 78.0679609



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.37868, 77.89498



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.37850, 77.89453



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.16764, 78.01565



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.15573, 78.02176



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.14545, 78.02208



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.14480, 78.02291



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 10.96071, 78.06796



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.96066, 78.06791



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 10.96123, 78.06585

